SURFACE ENGINEERING
Wear Resistance Specialists

SP45 Data Sheet

Description: Surface Engineering SP45 is a nickel chrome boron alloy with a low melting point. It flows easily on to base metals and yields smooth deposits. It can be formed and wiped to a sharp edge. Deposits maintain a level of hardness up to 1200 F and resist galling, pitting and corrosion in addition to abrasion. It can be deposited on low and medium carbon steels and cast iron (with flux). Deposits have exceptional hot hardness capabilities.

Applications: Liners, Thrust Shoes, Bushings, Cages, Valve Components, Oil Extrusion Screws, Glass Molds, Centrifuges

Typical Chemical Analysis (Conforms to Chemical Composition of AWS 5.21, Class ENiCr-B)							
C	**Si**	**Ni**	**Cr**	**Fe**	**Co**	**B**	**TAO**
0.5	3.5	BAL	12	3	1	2.5	<0.50

Machineability: Grind, Machining not recommended

Typical Mechanical Properties:

Tensile Strength: 77 ksi, As Cast
Tensile Elongation: <1%
Density: 7.8 g/cc
Liquidus: 1000 degrees C, 1850 degrees F
Hardness: 45-50 Rc.

Product Forms:
Cast Welding Rods, Welding Wire , Powders, Cast Components

Note: The data is presented as typical for this alloy, actual specifications will vary from heat

SURFACE ENGINEERING
Wear Resistance Specialists

PRODUCT TESTING AND PROPERTIES

Product: SP88T HVOF Powder
Form: Powder
Size: 270/22µ

PTP No 59
Rev: 2 (Rev HF)
Date: 2/26/26

Specifications

None

Quality Tests Required		
Test	**Procedure**	**Frequency**
Chemistry per GD-OES	LECOGDS-001	Every heat
Hardness	HARDNESS-001	
Apparent Density	APPDENSITY-001	
Hall Flow	HALLFLOW-001	
Sieve Analysis	SIEVE-001	
Special Testing Instructions:		

Physical Properties			
	Hardness	**Hall Flow**	**Apparent Density**
270/22µ	58-64 HRC	17.0s/50g max	4 - 5 g/cm^3

Chemistry (% by weight)							
Button Specimen	**C**	**Si**	**Cr**	**Ni**	**W**	**Fe**	**B**
270/22µ	0.5-1.0	3.5-4.5	13-15	BAL	13.0-15.0	3.5-5.5	3.0-4.0

Sieve Analysis (% by weight)							
270/22µ	**+230**	**+270**	**+325**	**+400**	**+450**	**+500**	**Pan**
	1.0 max	BAL				5.0 max	1.0 max

Approval

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SURFACE ENGINEERING
Wear Resistance Specialists

PRODUCT TESTING AND PROPERTIES

Product: SP45 HVOF Powder
Form: Powder
Size: 270/22µ

PTP No 41
Rev: 3 (Rev HF)
Date: 2/25/26

Specifications

AWS A5.21, Class ERNiCr-B (Chemistry Only)

Quality Tests Required		
Test	**Procedure**	**Frequency**
Chemistry per GD-OES	LECOGDS-001	Every heat
Hardness	HARDNESS-001	
Apparent Density	APPDENSITY-001	
Hall Flow	HALLFLOW-001	
Sieve Analysis	SIEVE-001	
Special Testing Instructions:		

Physical Properties			
	Hardness	**Hall Flow**	**Apparent Density**
270/22µ	43-50 HRC	17.0s/50g max	4 - 5 g/cm^3

Chemistry (% by weight)						
Button Specimen	**C**	**Si**	**Cr**	**Ni**	**Fe**	**B**
270/22µ	0.45-0.55 Aim 0.50	2.8-3.5 Aim 3.0	10.0-13.0 Aim 11.5	BAL	3.5-4.8 Aim 4.0	2.5-3.0 Aim 2.7

Sieve Analysis (% by weight)						
270/22µ	**+230**	**+325**	**+400**	**+450**	**+500**	**Pan**
	1.0 max	BAL			5.0 max	1.0 max

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SURFACE ENGINEERING
Wear Resistance Specialists

PRODUCT TESTING AND PROPERTIES

Product: 80/20 NiCr Micro-Velocity HVOF Powder
Form: Powder
Size: 270/22µ

PTP No 15
Rev: 2 (Rev HF)
Date: 2/23/2026

Specifications

None

Quality Tests Required		
Test	**Procedure**	**Frequency**
Chemistry per GD-OES	LECOGDS-001	Every heat
Hardness	HARDNESS-001	
Apparent Density	APPDENSITY-001	
Hall Flow	HALLFLOW-001	
Sieve Analysis	SIEVE-001	
Special Testing Instructions:		

Physical Properties			
	Hardness	**Hall Flow**	**Apparent Density**
270/22µ	N/R	17s/50g max	4-5 g/cm^3

Chemistry (% by weight)						
Button Specimen	**C**	**Mn**	**Si**	**Cr**	**Ni**	**Fe**
270/22µ	Low as Possible	2.5 max	1.5 max	19.0-21.0	78.5-81.5	1.0 max

Sieve Analysis (% by weight)

270/22µ	**+270**	**+325**	**+400**	**+450**	**+500**	**+635**	**Pan**
	0.0 max	BAL					3.0 max

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SURFACE ENGINEERING
Wear Resistance Specialists

Cobalt 6 Data Sheet

Description: Cobalt alloy 6 is a Cobalt, high chromium, tungsten alloy. It is Excellent for corrosion resistance, wear and galling. It is the most flexible and widely used of the cobalt alloys due to overall performance. Cobalt 6 retains hardness up to 930°F (500°C)

Applications: Shear blades, Fluid Flow Valves, Extrusion Screws, Roll Bushings, High Temperature, Valve Bearing Surface

Typical Chemical Analysis:

C	Mn	Si	Ni	Cr	W	Mo	Fe	Co
1.1	0.1	1.3	2.5	28.2	4.7	0.1	2.5	BAL

Machinability: Good, Use carbide tools

Density: 0.303 lbs/in^3, 8.39 g/cc

Hardness: 35 - 48 HRc

Liquidus: 2360°F, 1292°C

Meets these Specifications: AWS/SFA A5.21 ERCoCr-A

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SURFACE ENGINEERING
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PRODUCT TESTING AND PROPERTIES

Product: SP77P HVOF Powder
Form: Powder
Size: 230/31µ

PTP No 16
Rev: 2 (Rev HF)
Date: 2/23/2026

Specifications

None

Quality Tests Required		
Test	**Procedure**	**Frequency**
Chemistry per GD-OES	LECOGDS-001	Every heat
Hardness	HARDNESS-001	
Apparent Density	APPDENSITY-001	
Hall Flow	HALLFLOW-001	
Sieve Analysis	SIEVE-001	
Special Testing Instructions:		

Physical Properties			
	Hardness	**Hall Flow**	**Apparent Density**
230/31µ	50-58 HRC	17s/50g max	4.5 g/cm^3 min.

Chemistry (% by weight)							
Button Specimen	**C**	**Si**	**Cr**	**Ni**	**Fe**	**P**	**B**
230/31µ	0.02 max	4.5-5.0	6.5-7.5	BAL	1.0 max	0.5-1.0	3.2-3.8

Sieve Analysis (% by weight)							
230/31µ	**+230**	**+270**	**+325**	**+400**	**+450**		**Pan**
	0 max	BAL					3 max

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SURFACE ENGINEERING
Wear Resistance Specialists

PRODUCT TESTING AND PROPERTIES

Product: SP46 HVOF Powder
Form: Powder
Size: 230/22µ

PTP No 7
Rev: 5 (rev C)
Date: 03/24/2026

Specifications

AWS A5.21, Class ERNiCr-C (Chemistry Only)

Quality Tests Required		
Test	**Procedure**	**Frequency**
Chemistry per GD-OES	LECOGDS-001	Every heat
Hardness	HARDNESS-001	
Apparent Density	APPDENSITY-001	
Hall Flow	HALLFLOW-001	
Sieve Analysis	SIEVE-001	
Special Testing Instructions:		

Physical Properties			
	Hardness	**Hall Flow**	**Apparent Density**
230/22µ	56-62	17.0s/50g max	4.0-5.0 g/cm^3

Chemistry (% by weight)						
Button Specimen	**C**	**Si**	**Cr**	**Ni**	**Fe**	**B**
230/22µ	0.50-1.00	3.5-4.5	16-18	BAL	3.0-5.5	3.0-3.5

Sieve Analysis (% by weight)

230/22µ	**+230**	**+270**	**+325**	**+400**	**+400**	**+500**	**Pan**
	1.0 max	BAL					5.0 max

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SURFACE ENGINEERING
Wear Resistance Specialists

PRODUCT TESTING AND PROPERTIES

Product: SP46SF Spray Fuse Powder
Form: Powder
Size: 200/400

PTP No 5
Rev: 2 (Rev C)
Date: 3/20/2026

Specifications

AWS A5.21, Class ERNiCr-C (Chemistry Only)

Quality Tests Required		
Test	**Procedure**	**Frequency**
Chemistry per GD-OES	LECOGDS-001	Every heat
Hardness	HARDNESS-001	
Apparent Density	APPDENSITY-001	
Hall Flow	HALLFLOW-001	
Sieve Analysis	SIEVE-001	
Spray Test	SPRAY-001	
Special Testing Instructions:		

Physical Properties			
	Hardness	**Hall Flow**	**Apparent Density**
200/400	56-62	14-17s/50g	4.0-5.0 g/cm^3

Chemistry (% by weight)						
Button Specimen	**C**	**Si**	**Cr**	**Ni**	**Fe**	**B**
200/400	0.50-1.00 (aim 0.65)	3.5-4.5 (aim 3.7)	14.0-16.0 (aim 15.0)	BAL	3.0-5.5 (aim 3.0)	3.0-4.0 (aim 3.6)

Sieve Analysis (% by weight)

200/400	**+170**	**+200**	**+230**	**+270**	**+325**	**+400**	**Pan**
	1.0 max	BAL				15.0 max	3.0 max

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